UGANDA CHRISTIAN UNIVERSITY
A Centre of Excellence in the Heart of Africa

FACULTY OF ENGINEERING, DESIGN AND TECHNOLOGY
DEPARTMENT OF COMPUTING AND TECHNOLOGY
EASTER 2026 SEMESTER EXAMINATION

PROGRAM: BACHELOR OF SCIENCE IN COMPUTER SCIENCE
BACHELOR OF SCIENCE IN DATA SCIENCE AND ANALYTICS

YEAR: 2

SEMESTER: 2

COURSE CODE: CSC1204

COURSE NAME: MACHINE LEARNING

EXAMINATION TYPE: PROJECT-BASED EXAM

EXAMINATION DATE: APRIL 2026

TIME ALLOWED: 2 WEEKS

Examination Instructions

1. The general Uganda Christian University examination guidelines and academic & financial policies apply to this examination. Violating any of the policies by the student automatically makes this examination attempt void, even if you have completed and submitted the answer booklet.
2. This is a project-based examination worth 100 marks,
   i. This is a project-based examination worth 100 marks. Students must complete ALL sections as described in this paper.
   ii. All implementation must be done in Python using a Jupyter Notebook (`.ipynb`). The notebook must be well-organized with clear section headings, markdown explanations, and fully executable code.
   iii. Students must also submit a written project report (3–4 pages, excluding references and appendices) in LaTeX PDF format. The report must follow the structure outlined in Section E of this paper, and all submissions must be made on the Moodle platform.
3. Uganda Christian University or its officials reserve the right of admission or disqualification from this exam.

Section A: Problem Definition & Dataset Acquisition [15 Marks]

In this section, you will identify a real-world problem that can be addressed using machine learning techniques and acquire an appropriate dataset to support your investigation.

(a) **Problem Statement** **[5 marks]**

(i) Clearly define a real-world problem that you intend to solve using machine learning. State the problem in 1–2 paragraphs. Your problem must involve either:

- A supervised learning task (classification or regression), OR
- An unsupervised learning task (clustering or dimensionality reduction), OR
- A neural network / perceptron-based approach with gradient descent.

(ii) Explain *why* this problem matters. Who are the stakeholders? What decisions could be improved by a machine learning solution?

(iii) State your project objective(s) clearly.

(b) **Dataset Acquisition** **[5 marks]**

(i) Identify and obtain a suitable dataset from a reputable source.

(ii) The dataset must have:

- At least 200 records (rows).
- At least 5 features (columns), with a mix of numeric and/or categorical variables.
- A clearly identifiable target variable (for supervised tasks) or meaningful features suitable for clustering (for unsupervised tasks).

(iii) Provide the full citation and URL of the dataset source.

(c) **Data Description** **[5 marks]**

(i) In your notebook, load the dataset using Pandas and display the first 10 rows.

(ii) Report the shape of the dataset (rows $\times$ columns), all column names, and data types.

(iii) Provide a brief description of each column (a data dictionary table is expected).

(iv) Check for missing values and report the count per column.

Section B: Exploratory Data Analysis (EDA) [20 Marks]

Before building any model, you must thoroughly explore and understand the data. This section tests your ability to summarize, visualize, and prepare the data for modelling.

(a) **Summary Statistics & Data Cleaning** **[6 marks]**

- (i) Compute descriptive statistics for all numeric features. Present the results in a table.
- (ii) Handle any missing values using an appropriate strategy. Justify your choice.
- (iii) Identify and handle any outliers or anomalous values. Explain your approach.
- (iv) If the dataset contains categorical variables, encode them using label encoding or one-hot encoding as appropriate. Explain your choice.

(b) **Univariate Analysis** **[4 marks]**

- (i) Plot the distribution of the target variable. Comment on whether the data is balanced or skewed.
- (ii) Plot histograms or boxplots for at least 3 important numeric features. Discuss any notable patterns.

(c) **Bivariate and Multivariate Analysis** **[6 marks]**

- (i) Generate a correlation heatmap for all numeric features. Identify the features most strongly correlated with the target variable (or with each other, for unsupervised tasks).
- (ii) Create at least 2 scatter plots or grouped visualizations that explore relationships between key features and the target.
- (iii) Produce a pairplot for the most relevant features.

(d) **Key Insights** **[4 marks]**

- (i) Summarize your EDA findings in 3–5 bullet points (in a markdown cell). These should highlight patterns, relationships, or data quality issues that will influence your modelling decisions.
- (ii) Based on your EDA, state which features you expect to be most important for your model and why.

Section C: Model Building & Training [25 Marks]

This is the core technical section of the project. You will build, train, and tune at least two machine learning models appropriate for your chosen problem.

(a) **Data Preparation** **[5 marks]**

(i) Identify the feature matrix (X) and the target vector (y). For unsupervised tasks, identify the feature matrix used for clustering.

(ii) Split the data into training (80%) and testing (20%) sets using `random_state=42`. For unsupervised tasks, explain how you will validate cluster quality instead of a train/test split.

(iii) Apply feature scaling where necessary. Explain *why* scaling is or is not needed for your chosen algorithm(s).

(b) **Model 1: Primary Model** **[10 marks]**

(i) Choose and justify a primary machine learning algorithm from the list below:

Approach	Algorithms
Supervised, Classification	KNN, Decision Tree, Random Forest, Logistic Regression, SVM
Supervised, Regression	Linear Regression, Random Forest Regressor, SVR
Unsupervised	K-Means, Hierarchical Clustering, DBSCAN
Neural Network	Perceptron, Multi-Layer Perceptron with Gradient Descent

(ii) Train the model on the training set.

(iii) Perform hyperparameter tuning by experimenting with at least **3 different configurations** of the model's key hyperparameters. Record the performance metric for each configuration.

(iv) Explain the role of each hyperparameter you tuned and how it affects the model's behaviour.

(c) **Model 2: Comparison Model** **[10 marks]**

(i) Choose a second, different algorithm from the table above. Justify your choice.

(ii) Train this model on the same training set.

(iii) Perform hyperparameter tuning with at least 2 different configurations.

(iv) Record performance metrics for each configuration.

Section D: Model Evaluation & Comparison [20 Marks]

A well-trained model means nothing without rigorous evaluation. In this section, you will evaluate and compare the performance of your models using appropriate metrics and visualizations.

(a) **Evaluation Metrics** **[8 marks]**

Depending on your task type, apply the appropriate evaluation metrics:

Task Type	Required Metrics
Classification	Accuracy, Precision, Recall, F1-Score, Confusion Matrix
Regression	MAE, MSE, RMSE, R^2 Score
Clustering	Silhouette Score, Within-Cluster Sum of Squares (WCSS)

(i) For classification: Display the confusion matrix as a heatmap and print the full classification report (precision, recall, F1-score per class) for the best configuration of each model.

(ii) For regression: Report MAE, MSE, RMSE, and R^2 for the best configuration of each model. Create a scatter plot of predicted vs. actual values with a reference line ($y = x$).

(iii) For clustering: Report the silhouette score and WCSS. Create a 2D scatter plot of clusters with centroids marked.

(b) **Model Comparison** **[7 marks]**

(i) Create a comparison table summarizing the key performance metrics of all model configurations tested.

(ii) Identify the best-performing model and configuration. Justify your choice based on the evaluation metrics.

(iii) Discuss the trade-offs between the two models.

(c) **Visualizations** **[5 marks]**

(i) Produce at least two evaluation-related visualizations. Examples include:

- Confusion matrix heatmap
- Predicted vs. actual scatter plot
- Residual plot
- Feature importance bar chart
- Elbow curve or silhouette plot
- Decision tree visualization

(ii) Each visualization must have a title, axis labels, and a brief interpretation in a markdown cell.

Section E: Project Report [20 Marks]

You are required to write a concise project report of **3–4 pages** (excluding references and appendices) summarizing your entire project. The report must be submitted as a PDF document and must follow the structure below.

(a) **Report Structure** **[15 marks]**

The report must contain the following sections:

(i) Introduction **[3 marks]**
State the problem, its significance, and the objectives of the project. Briefly describe the dataset used and its source.

(ii) Methodology **[4 marks]**
Describe the data science lifecycle steps you followed: data collection, preprocessing, EDA, feature engineering, model selection, training, and evaluation. Mention the tools and libraries used.

(iii) Results **[4 marks]**
Present the key findings from your EDA and model evaluation. Include at least one table and one figure. State the performance of your best model clearly.

(iv) Discussion & Conclusion **[3 marks]**
Interpret the results. What did the model learn? What are the limitations of your approach? Suggest possible improvements or future work.

(v) References **[1 mark]**
Cite all sources used, including the dataset, any tutorials, documentation, or academic papers. Use a consistent citation style.

(b) **Report Quality** **[5 marks]**

(i) Writing is clear, grammatically correct, and free of spelling errors.

(ii) The report is well-organized with clear headings and logical flow.

(iii) Tables and figures are properly numbered, captioned, and referenced in the text.

(iv) Technical terms are used correctly and explanations are accessible.

(v) The report adheres to the 3–4 page limit.

Project Presentation and Defense

Each student is required to present and defend their project before a panel. The presentation is a mandatory component of this examination. Failure to present will result in an incomplete grade.

1. Each student will be allocated 10 minutes to present their project, followed by 5 minutes of questions from the panel.
2. Students must demonstrate a clear understanding of their own code and be able to explain any part of the notebook when asked.
3. The presentation will be assessed as part of the Project Report marks in Section E. Students who present well and demonstrate strong understanding of their work will be awarded full marks in the report quality component.

Grading Rubric [100 Marks Total]

The following rubric summarizes how each component of the examination will be assessed.

Area	Description	Marks
Section A: Problem Definition & Dataset Acquisition		
Problem Statement	Clear definition of a real-world ML problem; significance and stakeholders explained; objectives well stated.	5
Dataset Acquisition	Suitable dataset from a reputable source; meets minimum requirements (200+ rows, 5+ features, identifiable target); proper citation and URL provided.	5
Data Description	Dataset loaded and displayed correctly; shape, column names, and data types reported; data dictionary provided; missing values checked and reported.	5
Section B: Exploratory Data Analysis (EDA)		
Summary Statistics & Data Cleaning	Descriptive statistics computed and presented; missing values handled with justification; outliers identified and addressed; categorical encoding applied and explained.	6
Univariate Analysis	Target variable distribution plotted with commentary on balance/skew; histograms or boxplots for at least 3 numeric features with discussion.	4
Bivariate & Multivariate Analysis	Correlation heatmap generated with key correlations identified; at least 2 scatter plots or grouped visualizations; pairplot produced for relevant features.	6
Key Insights	3–5 bullet-point EDA summary provided; important features identified with reasoning linked to EDA findings.	4
Section C: Model Building & Training		
Data Preparation	Feature matrix and target vector correctly identified; proper train/test split (80/20, random_state=42); feature scaling applied or justified as unnecessary.	5
Model 1: Primary Model	Algorithm justified; model trained correctly; at least 3 hyperparameter configurations tested with results recorded; role of each hyperparameter explained.	10
Model 2: Comparison Model	Second algorithm justified; model trained on same data; at least 2 hyperparameter configurations tested with results recorded.	10
Section D: Model Evaluation & Comparison		

Area	Description	Marks
Evaluation Metrics	Correct metrics applied for task type; confusion matrix / regression plots / cluster plots presented; classification report or metric values clearly stated.	8
Model Comparison	Comparison table of all configurations; best model identified with metric-based justification; trade-offs between models discussed.	7
Visualizations	At least 2 evaluation visualizations produced; each has title, axis labels, and markdown interpretation.	5
Section E: Project Report		
Introduction	Problem, significance, objectives, and dataset described clearly.	3
Methodology	Data science lifecycle steps described; tools and libraries mentioned; process is reproducible.	4
Results	Key EDA and evaluation findings presented; at least one table and one figure included; best model performance stated.	4
Discussion & Conclusion	Results interpreted meaningfully; limitations acknowledged; future improvements suggested.	3
References	All sources cited; consistent citation style used.	1
Report Quality	Clear writing; well-organized; tables/figures numbered and captioned; correct technical language; 3–4 page limit adhered to.	5
	TOTAL	**100**

◀ END ▶